Ateneo de Manila University

Archīum Ateneo

Ateneo School of Medicine and Public Health Publications

Ateneo School of Medicine and Public Health

10-1-2023

Socio-Economic Factors and Its Influence on the Association Between Temperature and Dengue Incidence in 61 Provinces of the Philippines, 2010–2019

Xerxes Seposo
Hokkaido University

Sary Valenzuela
Ateneo de Manila University, rfvalenzuela@ateneo.edu

Geminn Louis Apostol
Ateneo de Manila University, gapostol@ateneo.edu

Follow this and additional works at: https://archium.ateneo.edu/asmph-pubs

 Part of the Tropical Medicine Commons

Custom Citation

Seposo X, Valenzuela S, Apostol GL (2023) Socio-economic factors and its influence on the association between temperature and dengue incidence in 61 Provinces of the Philippines, 2010–2019. PLoS Negl Trop Dis 17(10): e0011700. https://doi.org/10.1371/journal.pntd.0011700

This Article is brought to you for free and open access by the Ateneo School of Medicine and Public Health at Archīum Ateneo. It has been accepted for inclusion in Ateneo School of Medicine and Public Health Publications by an authorized administrator of Archīum Ateneo. For more information, please contact oadrcw.ls@ateneo.edu.

PLOS NEGLECTED TROPICAL DISEASES

RESEARCH ARTICLE

Socio-economic factors and its influence on the association between temperature and dengue incidence in 61 Provinces of the Philippines, 2010–2019

Xerxes Seposo[1,2,3]*, **Sary Valenzuela**[3], **Geminn Louis Apostol**[3]

1 Department of Hygiene, Hokkaido University, Sapporo, Hokkaido Japan, **2** School of Tropical Medicine and Global Health, Nagasaki University, Nagasaki, Japan, **3** Ateneo School of Medicine and Public Health, Ateneo de Manila University, Pasig, Philippines

* seposo.xerxestesoro@pop.med.hokudai.ac.jp

Check for updates

OPEN ACCESS

Citation: Seposo X, Valenzuela S, Apostol GL (2023) Socio-economic factors and its influence on the association between temperature and dengue incidence in 61 Provinces of the Philippines, 2010–2019. PLoS Negl Trop Dis 17(10): e0011700. https://doi.org/10.1371/journal.pntd.0011700

Editor: Qu Cheng, Huazhong University of Science and Technology Tongji Medical College, CHINA

Received: April 14, 2023

Accepted: October 4, 2023

Published: October 23, 2023

Peer Review History: PLOS recognizes the benefits of transparency in the peer review process; therefore, we enable the publication of all of the content of peer review and author responses alongside final, published articles. The editorial history of this article is available here: https://doi.org/10.1371/journal.pntd.0011700

Copyright: © 2023 Seposo et al. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

Data Availability Statement: The data set for dengue incidence was acquired with permission from the Philippines Department of Health Epidemiological Bureau, while that of the provincial

Abstract

Background

Temperature has a significant impact on dengue incidence, however, changes on the temperature-dengue relationship across axes of socio-economic vulnerability is not well described. This study sought to determine the association between dengue and temperature in multiple locations in the Philippines and explore the effect modification by socio-economic factors.

Method

Nationwide dengue cases per province from 2010 to 2019 and data on temperature were obtained from the Philippines' Department of Health–Epidemiological Bureau and ERA5-land, respectively. A generalized additive mixed model (GAMM) with a distributed lag non-linear model was utilized to examine the association between temperature and dengue incidence. We further implemented an interaction analysis in determining how socio-economic factors modify the association. All analyses were implemented using R programming.

Results

Nationwide temperature-dengue risk function was noted to depict an inverted U-shaped pattern. Dengue risk increased linearly alongside increasing mean temperature from 15.8 degrees Celsius and peaking at 27.5 degrees Celsius before declining. However, province-specific analyses revealed significant heterogeneity. Socio-economic factors had varying impact on the temperature-dengue association. Provinces with high population density, less people in urban areas with larger household size, high poverty incidence, higher health spending per capita, and in lower latitudes were noted to exhibit statistically higher dengue risk compared to their counterparts at the upper temperature range.

mean weekly temperatures were acquired from the Philippine Atmospheric, Geophysical and Astronomical Services Administration. The data sets may be accessed upon formal request by contacting the data source owners directly at nec.doh@gmail.com +63 (02) 8651 7800 and (+632) 8927-1541 or information@pagasa.dost.gov.ph, respectively. R codes to perform the analyses are available upon formal request from the Department of Hygiene, Graduate School of Medicine, Hokkaido University at eisei@med.hokudai.ac.jp.

Funding: XTS was supported by the Japan Society for the Promotion of Science Project number 22K17374. XTS and GLA were supported by Ateneo de Manila University URC 2022-18. The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

Competing interests: The authors have declared that no competing interests exist.

Conclusions

This observational study found that temperature was associated with dengue incidence, and that this association is more apparent in locations with high population density, less people in urban areas with larger household size, high poverty incidence, higher health spending per capita, and in lower latitudes. Differences with socio-economic conditions is linked with dengue risk. This highlights the need to develop interventions tailor-fit to local conditions.

Author summary

This study examined the effect of temperature on dengue in the 61 Provinces of the Philippines. We found that as temperature increases, so does the risk of dengue. However, the increase in the risk of dengue plateaus at 27.5 degrees Celsius, and beyond that temperature the risk decreases. The association between temperature and dengue incidence at high temperatures was stronger among provinces with high population density, less people in urban areas with larger household size, high poverty incidence, higher health spending per capita, and in lower latitudes compared to their counterparts. In particular, results showed that people in the rural areas had higher dengue risk than those in urban areas, which is contrary to expectations that dengue is an urban-dwelling related disease. On another hand, we found that high health spending per capita does not equate to better health outcomes. Efficiency of use of resources may come into play in terms of improving the health outcomes and not just the quantity or amount of the difference in the dengue risk by socio-economic factors highlights the need for a tailor-fit dengue response.

Introduction

Dengue, transmitted by the mosquito vector *Aedes aegypti*, remains a major public health concern in tropical areas [1]. Over the last 50 years, dengue incidence has increased globally by 30-fold and 70% of the global burden of disease comes from Asia [2]. In 2019, the Philippines declared dengue a national epidemic of concern with 420,453 cases. Similar increases in dengue morbidity have been noted in Vietnam and Malaysia [3]. Heightened dengue transmission may be partly driven by the increase in international travel, unplanned urbanization, lack of effective vector management, climate change, and poor socio-economic status, among others [1,4].

Literature identifies ambient temperature as an important meteorological risk factor for dengue transmission [5]. A systematic review by Li et. al. [5] revealed that the minimum temperature of 18 degrees Celsius is a critical limiting risk factor for dengue transmission, with high ambient temperatures promoting vector proliferation. Other global meta-analyses pose that dengue epidemics have a critical range of 22–29 degrees Celsius and that extremely higher temperatures reduce vector survival time [1,6]. In the advent of climate change, several studies have shown that current projections of 2 degrees Celsius by the end of the next century will result in an expansion of latitude and altitude range and duration of transmission of dengue [5,7,8] or even a shift in geographical regions affected by vector-borne disease outbreaks [9].

There have been similar climate change-dengue studies conducted in Southeast Asia, observing an increase of dengue risk within a specified range of high temperatures. In

Thailand, vector efficiency increased in temperatures from 20–30 degrees Celsius, affecting annual cyclic patterns of dengue hemorrhagic fever epidemics in Bangkok [10]. Similar results were noted with the studies in Singapore [11–13]. A study in China [14] observed an inverse-U shape association between temperature and dengue. One study in the Philippines observed significant interaction effects between seasons, temperature, and altitude on the hatch rates (HRs) of *A*. *aegypti* larvae and reproductive outputs (ROs) of mosquitoes [15]. HRs in both seasons were highest at 25 degrees Celsius and lowest at 38 degrees Celsius. ROs were highest at 25 degrees Celsius in the wet season and at 18 degrees Celsius in the dry season. Furthermore, the study noted that the incidence of the dengue-causing mosquito phenotype was found to be highest at 18 degrees Celsius and present even at 38 degrees Celsius in both seasons, suggesting larval adaptation to global warming [15].

Several temperature-dengue studies have documented evident heterogeneity in the association, which may potentially be due to socio-demographic and economic factors innate to the specific locations [16–19]. Certain vulnerable sub-populations have been observed to have higher infectious disease risk incidence such as those with higher poverty incidence, urban to population ratios, and smaller government health expenditures, as they have lower adaptive capacity and are more susceptible to negative health outcomes when exposed to environmental hazards [20–22]. While noted, examination of the contributors to heterogeneity of the linear association between temperature and dengue have been relatively scarce, with little to no evidence on how these socio-demographic and economic factors impact the non-linear temperature-dengue association. Against this backdrop, this study sought to determine the association between temperature and dengue in multiple locations in the Philippines, focusing on the role of socio-demographic and economic factors in the temperature-dengue association.

Methods

Ethics statement

The Ateneo de Manila University Research Ethics Committee waived the need for ethical approval and the need to obtain consent for collection, analysis, and publication of secondarily, retrospectively obtained for this non-interventional study. All methods were performed in accordance with the relevant guidelines and regulations.

Data source

Nationwide, weekly notifiable dengue cases per province (n = 81) from 2010–2019 were obtained from the Philippines' Department of Health–Epidemiological Bureau. In the Philippines, case surveillance is coursed through the Philippine Integrated Disease Surveillance and Response, which is subsequently managed by the Philippine Department of Health's Epidemiological Bureau. In this study, provinces (n = 61) with more than 300,000 population were utilized for analysis. Provinces with less than 300,000 population had a higher percentage of 0 observations, affecting statistical power and the subsequent uncertainty in the estimation, and were thus excluded from the current study.

In the absence of consistent background monitoring data for each of the provinces, hourly temperature was obtained from ERA5-land, which has a spatial resolution of 9km on a reduced Gaussian grid. Developed by the Copernicus Climate Change Service (C3S) on behalf of the European Union, ERA5-land is a publicly available enhanced global dataset that has shown accurate temperature readings in Europe and Central Asia [23–25]. Hourly temperatures were then aggregated to daily 24 hour mean temperature, and thereafter was aggregated into a weekly measure. Statistical correlation of temperature obtained from ERA5-land and those locations with existing background monitoring stations is high. For the case of Manila

City, Pearson's correlation coefficient was at 0.88 (p-val < 0.05), whereas for Cebu City it was at 0.86 (p-val < 0.05) (S1 Fig). Over the same period, daily temperature was aggregated by epidemiological week (EpiWeek) to match the dengue incidence data. On a similar note, several studies have utilized ERA5-land in examining the effect of temperature on specific health outcomes [26–28].

Potential effect modifiers were obtained from literature as shown in S1 Table. The socioeconomic factors included were poverty incidence, population density, people living in urban areas, number of people per household, health expenditure, and latitude per province, which were sourced from relevant literatures [20–22,29–34]. The socio-economic factors used were province specific. All were originally yearly data, except for latitude. These terms have been operationalized as shown in S2 Table.

Statistical analyses

We employed a generalized additive mixed model (GAMM) in examining the nationally representative temperature dengue association. Dengue data was assumed to follow a negative binomial distribution. Subsequently, we included a crossbasis term for temperature and adjusted for several covariates as shown in Eq 1.

$$Y_{t,i} \sim Negative\ binomial \tag{Eq1}$$

$$Y_{t,i} \sim \alpha + logpop + cbtemp + s(Epiweek, k = 4) + (1|Year) + (1|Province) + Y_{t-1,i} + \boldsymbol{\varepsilon}$$

$Y_{t,i}$ is the weekly dengue incidence in time (t) of province (i). The weekly dengue incidence is assumed to follow a negative binomial distribution, which is similar to previous studies [35]. α is the intercept. *logpop* is the log of the population, which is an offset of the changing population sizes [5]. *cbtemp* is a crossbasis function of the exposure and lag dimensions of temperature, with both dimensions parameterized with 4 degrees of freedom (df) using natural cubic splines, respectively. The maximum lag was set at 18 weeks (or 4.5 months), based on the maximum lag sensitivity analyses (S2 Fig) and loosely similar to other studies whose maximum lag was set around 20 weeks [13,36]. After further examination, a seasonal pattern can be noted in Fig 1, with a peak of the dengue cases around 32nd/33rd week. Thus, we adjusted for the seasonality of dengue cases through a 4 df, based on a sensitivity analysis in S3 Fig. Owing to the variations in the annual distribution of cases per year (S4 Fig), we added a random effect term of year; (1|Year). A random effect term of Province (1|Province) was also added to emulate a (partial) pooling effect in estimating the nationally representative association. Several studies have noted the potential immunity in the population, which could inflate the susceptible population in the subsequent time scale. To model this phenomenon, Imai et al. [37] utilized the previous time scale's case as a potential adjustment. Here, we adjusted for the previous week's case; $Y_{t-1,i}$. $\boldsymbol{\varepsilon}$ is the error term.

Effect modification by socioeconomic factors was also examined through an interaction analysis. In brief, the 5th and 95th percentile values for the specific effect modifier represented the low and high percentiles. The respective values of the effect modifier were then deducted from the 5th or the 95th percentile values, which effectively centers these values around the low and high percentiles. The temperature crossbasis term is then multiplied by the centered effect modifier in either (low or high) percentiles. The resultant crossbasis represents the exposure-lag-response dependency corresponding to the percentile centering point of the effect

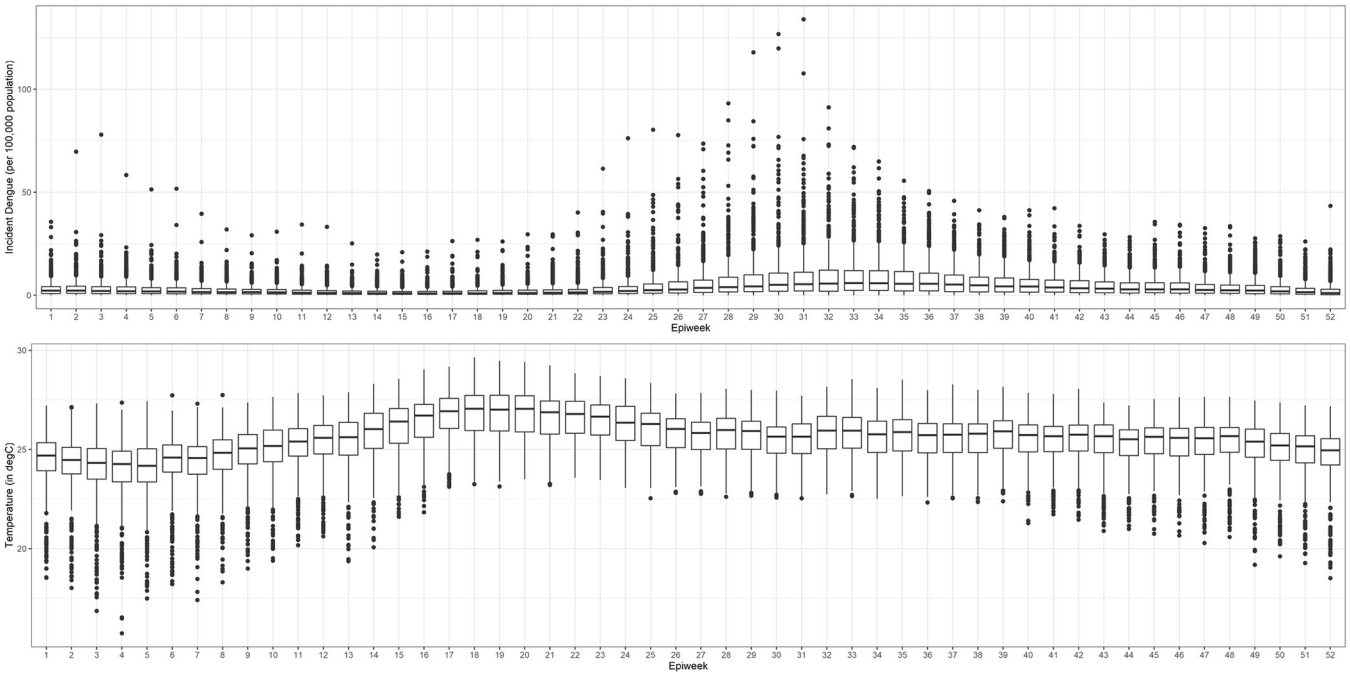


Fig 1. Boxplot diagrams of average dengue incidence (upper panel) and average recorded temperatures (lower panel) occurring per week in a year in the Philippines.

https://doi.org/10.1371/journal.pntd.0011700.g001

modifier [38].

$$Y_{t,i} \sim \alpha + logpop + cbtemp + effmod_j + cbtemp : effmod_j + s(Epiweek, k = 4) + (1|Year) + (1|Province) + Y_{t-1,i} + \varepsilon \quad \text{(Eq2)}$$

Here, $effmod_j$ represents the respective effect modifier centered at either percentile j (5th or 95th). All analyses were implemented using R statistical programming version 4.1.3 through the following packages: "dlnm" [39], "mixmeta" [40], "mgcv" [41], "lubridate" [42], and "dplyr" [43].

Results

Descriptive analyses

A mean, weekly dengue incidence of 2.128 was recorded from 2010–2019 across 61 provinces in the Philippines, with a range from 0 to 133.925 per EpiWeek. Recorded temperatures each week in each province had a median of 25.62 and a range of 15.73–29.64 per EpiWeek. The population per province varied greatly from 29,2781–13,484,462 people and a median of 89,0451 (Table 1) per EpiWeek.

Fig 2A shows the distribution of dengue incidence per 100,000 population from 2010–2019, in all study provinces across the three major island groups of Luzon (north), Visayas (middle), and Mindanao (south). The top three provinces recording the highest dengue incidence were the highly urbanized provinces of Davao del Sur (11.14 per 100,000 population), South Cotabato (8.19 per 100,000 population), and Misamis Oriental (6.61 per 100,000 population). Provinces reporting the lowest dengue incidence were Lanao del Sur (0.29 per 100,000 population), Tawi-tawi (0.26 per 100,000 population, and Sulu (0.10 per 100,000 population), which are high-conflict areas in Mindanao [44,45]. Fig 2B indicates the average reported temperature per

Table 1. Descriptive summary statistics for dengue cases, temperature, population, and dengue incidence across the 61 Provinces in the Philippines from 2010–2019.

	Median	Percentiles				Interquartile Range (IQR)	Range
		1st	25th	75th	99th		
Dengue Cases	23	0	7	64	577.81	57	0–2,748
Temperature (˚C)	25.62	20.92	24.64	26.45	28.23	1.78	15.73–29.64
Population	89,0451	308,985	617,333	1,697,050	12,877,253	1,079,717	29,2781–13,484,462
Dengue Incidence	2.128	0	0.787	5.290	31.180	4.503	0–133.925

https://doi.org/10.1371/journal.pntd.0011700.t001

province in the same time period. The provinces that reported the highest temperatures between 26–27 degrees Celsius clustered around central and southern Luzon, Visayas, and the northern areas of Mindanao. The highest recorded daily mean temperatures were in Tawi-tawi (27.04 degrees Celsius), Sulu (26.88 degrees Celsius), and Basilan (26.67 degrees Celsius).

Seasonal trend in dengue incidence is apparent (as shown in Fig 1, Upper Panel) with the gradual increase starting in EpiWeek 19 (1.05 per 100,000 population), peaks in EpiWeek 36 (5.56 per 100,000 population), then steadily decreases until the end of the year. In the lower panel, the range of the median value of the average temperature is around 23 to 27 degrees Celsius, peaking around EpiWeek 20 (26.78 degrees Celsius).

Association between temperature and dengue incidence

Fig 3 depicts the overall exposure-response association. Dengue-related risk is linearly increasing from the minimum temperature of 15.8 degrees Celsius and peaking at 27.5 degrees Celsius before declining. The inverted U-shaped risk pattern between temperature and dengue remains to be robust even after adjusting for precipitation as shown in S5 Fig. On the other hand, cumulative lag-response association at the 90th percentile, relative to the 50th percentile of the prediction interval, shows a reduced risk at the earlier lags but with a gradual increase in latter lags peaking

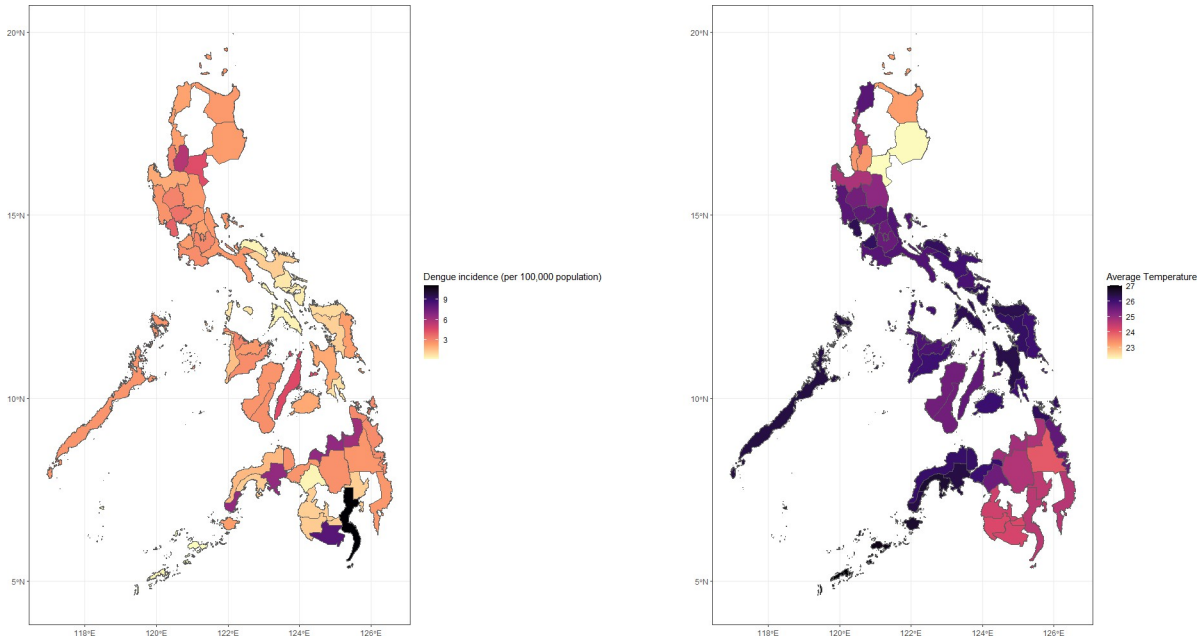


Fig 2. Heat map* of dengue incidence cases (left) and average temperature (right) per region in the Philippines from 2010–2019. * **Fig 2 is created using R package "ggmap".**

https://doi.org/10.1371/journal.pntd.0011700.g002

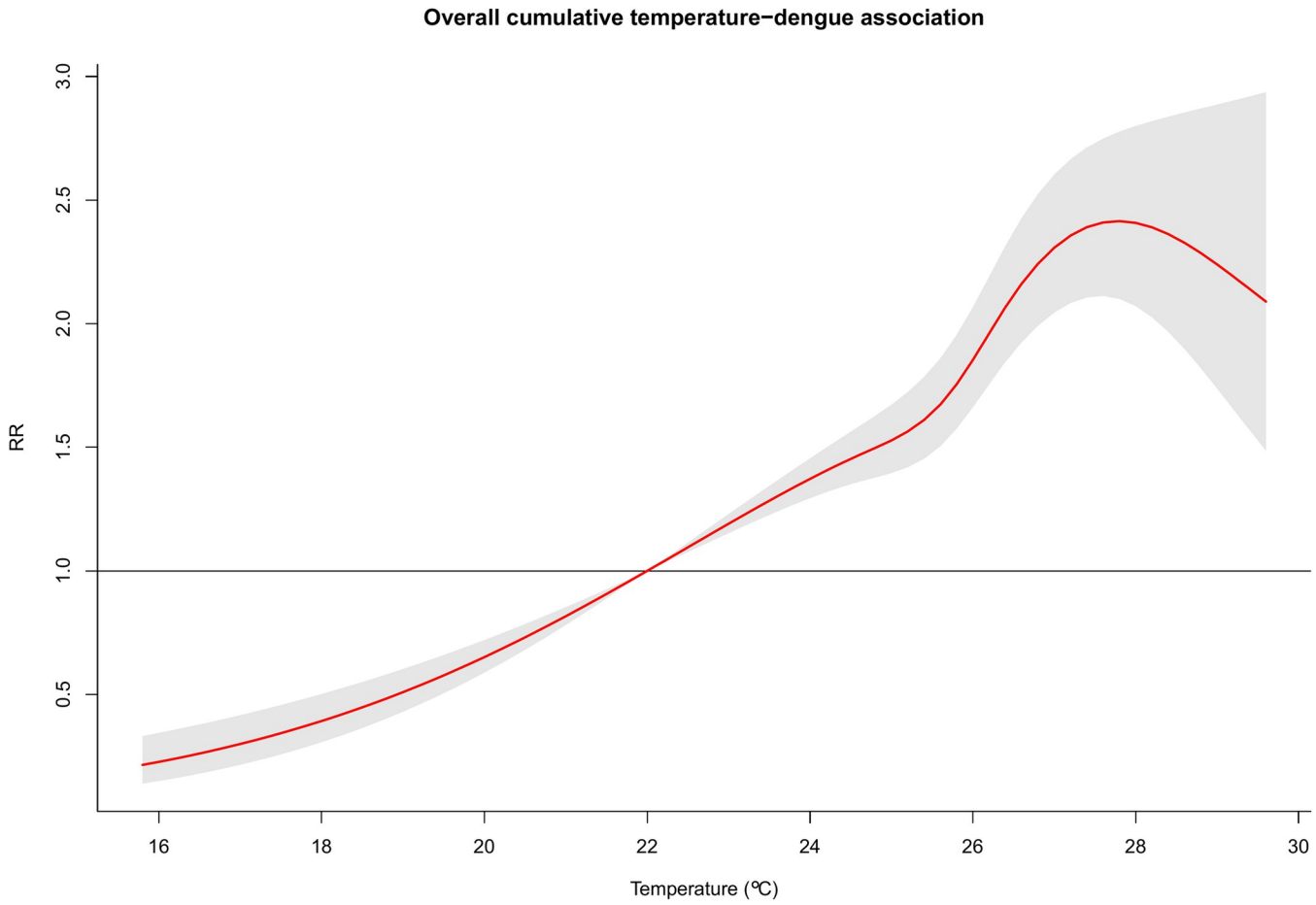


Fig 3. Nationally representative exposure-response function of increasing temperature with dengue incidence.

https://doi.org/10.1371/journal.pntd.0011700.g003

at around 16 weeks (or approximately 4 months), followed by a subsequent reduction in the risks (Fig 4). Further examination of the exposure- and lag-response associations by specific lag period (Weeks 0, 3, 6, 9, 12 and 18) and temperature percentiles (1st, 5th, 10th, 90th, 95th, and 99th), revealed nearly consistent patterns with the overall exposure and lag associations (in S6 Fig).

Effect modification by socio-economic factors

Socio-economic factors had varying impact on the temperature-dengue association (in Fig 5). Operationalization of the low and high for each socioeconomic variable is relative to each socioeconomic variable's 5th and 95th percentiles, respectively. Provinces with high population density, less people in urban areas with larger household size, high poverty incidence, higher health spending per capita, and in lower latitudes were noted to exhibit statistically higher dengue risk compared to their counterparts at the upper temperature range (Figs 5 and S7). Modifications in the lower temperature range were also observed for all effect modifiers, except for latitude. Main effects of the effect modifiers are shown in S3 Table.

Discussion

Nationally representative exposure-response function between temperature and dengue in the Philippines was noted to have an inverse U-shaped, non-linear association, with the relative

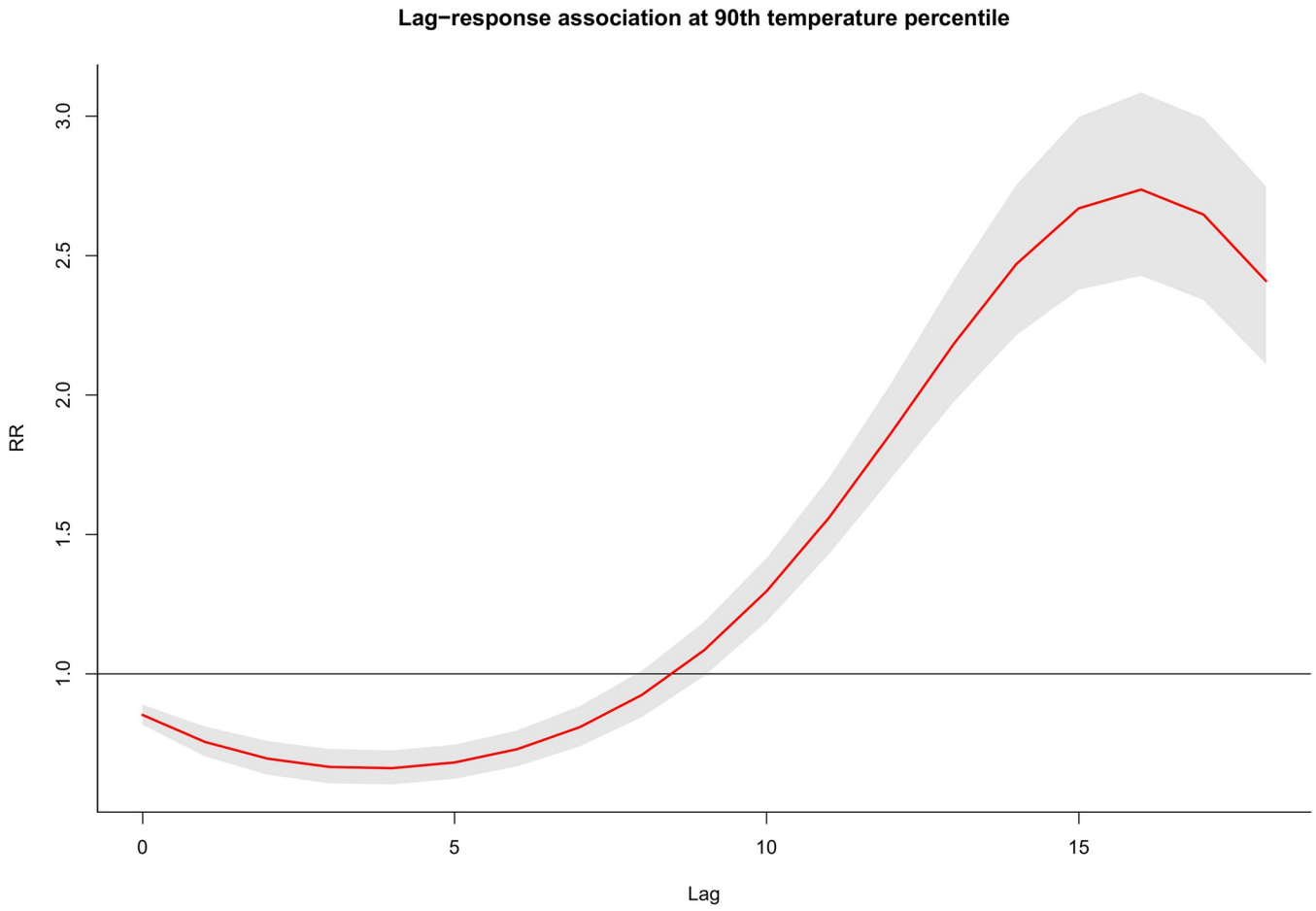


Fig 4. Nationally representative lagged temperature-dengue association at the 90th temperature percentile relative to the 50th percentile of the prediction interval.

https://doi.org/10.1371/journal.pntd.0011700.g004

risk peaking at 27.5 degrees Celsius. In Guangzhou, China, both Wu et al. [14] and Xiang et al. [46], also noted an inverse U-shaped temperature-dengue association. The decline in the risk beyond 27.5 degrees Celsius can be partially explained by the physiological mechanisms affecting vector survival. Extremely hot temperatures beyond maximum temperatures have an inverse effect on fecundity, blood size meal, and vector viability, reducing mosquito survival time [6,47]. Dengue fever risk was seen to increase after extremely high temperatures due to accelerated incubation and increased mosquito population, although dengue lifespan and egg survival rate was decreased [48]. Hawley [49] observed that the optimal thermal conditions for *A*. *albopictus* survival is around 20 to 27 degrees Celsius. Another study by Servadio et. al. [9,13], presented similar results of a similar parabolic curve, with maximum development rate in vector-borne diseases at temperatures between 28 and 32˚C and none under 18C or over 34˚C in South and Southeast Asian countries. This is further supported by other studies [50,51], which observed a decline in vector's survival rate beyond 28 degrees Celsius.

Our results have shown a 16-week (or approximately 4 months) lag association between temperature and dengue incidence as shown in Fig 4. Previous studies have also utilized longer lag periods to examine the lag association. In Singapore, Hii et al. [13] utilized a 20-week maximum lag to examine the effects of weather variables (including temperature) on dengue incidence. Results showed higher relative risks of dengue cases at a time lag of 3–4 months, which

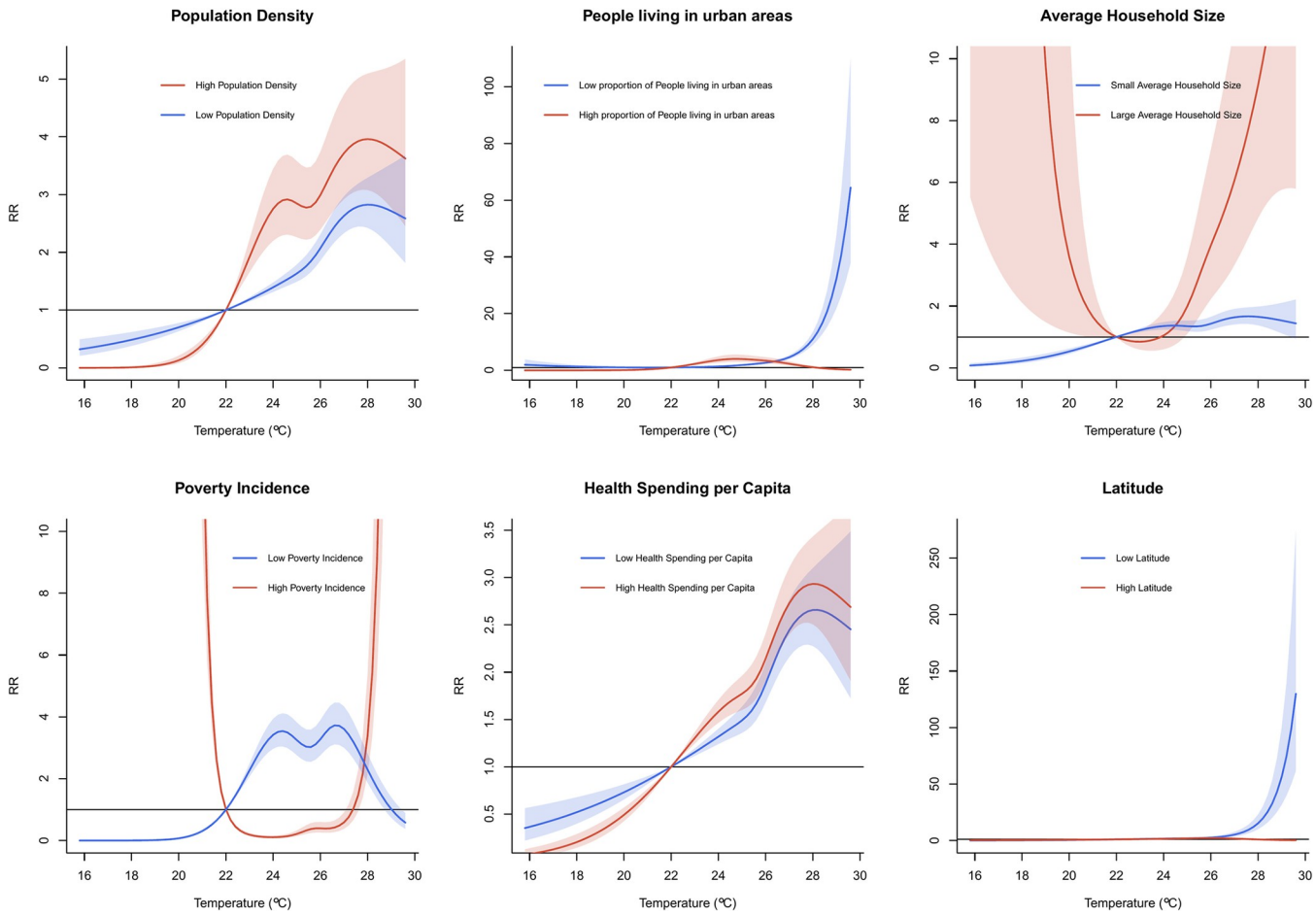


Fig 5. Effect modification of the temperature-dengue association by socio-economic variables.

https://doi.org/10.1371/journal.pntd.0011700.g005

is consistent with our study findings. Similarly, Hii et al. [36] also utilized a 20-week maximum lag in examining the optimal lead time for dengue forecasting (parameterized by weather variables), with an optimal lead time at around 3 months. Koh et al. [52] also found that dengue incidence in 2005 was highly associated with weekly mean temperature at a lag time of 18 weeks. Heng et al. [53] noted that dengue incidence occurred at a lag time of 8–20 weeks following elevated temperature. Similar 3-month lag associations have been observed in Bangkok [54] and Taiwan [14]. The delayed association between temperature and dengue incidence may potentially be related to the incubation periods in host–vector–pathogen transmissions cycle plus reproduction, maturation and survival rate of vector [13].

Similar to previous studies, higher population density was associated with a visually higher temperature-dengueran association than low population dense areas [33,55]. In Indonesia, Nuranisa, et al. [56] noted a positive correlational association between dengue incidence and population density. These highly dense areas provide suitable conditions for vector proliferation and subsequent disease transmission [57]. Noteworthily, urbanization revealed a reduced risk for temperature-related dengue incidence, contrary to prior literature, which associates them as well-documented risk factors [17,19,54,58–61]. Unplanned urbanization promotes close proximity to larval habitats, dengue-prone water storage practices, and rapid population movement [1,59]. Notably, the top three provinces for dengue incidence, Davao del Sur, South

Cotabato, and Misamis Oriental, are some of the more urbanized cities and more populated areas in the country [45]. These results suggest that the connection between dengue, human behavior, and temporal relationships is more complex than previously thought. As to whether dengue remains primarily a rural disease as a result of increased rural transmission or due to diminishing incidence rates in urban areas due to concentrated vector control efforts and increased health spending in urban areas, remains to be elucidated. Dengue transmission may indeed be increasing in rural areas due to the interplay of additional modifying variables that were not included in the study due to lack of geographical and behavioral data, such as forest cover or the presence of vegetation, open water containers, and degree of intrahousehold crowding, which are known dengue risk factors [1,17,19,54,58,59,61,62]. One study in Vietnam [34] found that risk of dengue was higher in rural areas due to the lack of piped water supply. Another study in Rio De Janeiro noted that while highly-populated, highly-urbanized areas displayed increased dengue incidence, so did other areas with low human density but with medium to highly-vegetated neighborhoods [62]. Further investigations are required to prove these hypotheses. While our analysis provides hindsight on the role of province-specific characteristics and its influence on the relationship between dengue and temperature, we cannot ascertain the possible mechanisms that are related to the observed associations.

Average household size was found to have higher temperature-related dengue risk than small households. Similar results were seen in New Caledonia [16]. Its known vectors, *Aedes aegypti* or *Aedes albopictus*, exhibit limited flight capacity and remain in close proximity to their breeding water containers [60,63]. Domestic workers and housewives are also more vulnerable to dengue compared to those who spend less time at home [17,59]. Larger families may also have larger dwellings, which may have more water-storage containers, such as flowerpots, vases, and lush areas that can serve as breeding sites [60]. As dengue cases have also been observed to cluster within households and other countries [63–65], intrahousehold dengue may be more prevalent than interhousehold dengue transmission. In contrast, a study in Brazil found large household sizes were protective against dengue risk, as smaller families (<7 people) were found to be living in more cramped spaces than their larger counterparts in the studied area [17]. This is in concurrence to the results observed by Zambrano, et. al. [59] who argued that the increased ratio of household members to the number of rooms, otherwise known as "crowding", was more predictive of dengue incidence than the absolute number of household members. Risks apparent in the lower temperature range among large household size families may potentially also be related to the persistence of disease transmission whose range falls between 21.6–32.9 degrees Celsius [46] coupled by the crowding of the household members.

Poverty incidence was also found to modify the temperature-dengue association in both low and high temperature ranges. Several studies associate poverty with dengue risk due to substandard housing, solid waste mismanagement, and sewerage systems that favor close proximity to larval habitats [17,58,59]. Mulligan et. al. [61] argues that dengue's epidemiological and transmission characteristics in relation to poverty should be interpretated in the context of how these specific poverty-related factors (i.e. household overcrowding, absence of air conditioning, low education, low income, poor housing quality, self-assessment of economic situation as poor, low family income, higher population density, and inappropriate use of covers on water containers, among others) affect the biology and ecology of the vector *Aedes aegypti*. In addition, poverty incidence may simply represent a variety of behavioral and exposure differences of the population. Resource-poor households/populations tend to work mostly during extreme conditions, including that during low temperatures and high temperatures in contrast to those resource-rich households/populations. Populations in resource-poor locations would prioritize more food and other basic necessities than personal risk prevention of dengue (e.g.

mosquito repelling lotion, long sleeves, etc.). Since dengue transmission may persist in a range of 21.6–32.9 degrees Celsius [46]. If we chart against Fig 4 (poverty incidence panel), we note that the range 21 to 29 degrees Celsius, which is well within the range of dengue transmission, hence the potential high risk at these extremes for the high poverty incidence populations. The low risk at the middle may also be similarly related to low exposure of the population since these are temperatures that are conducive for most less labor-intensive jobs, whereby these populations don't engage with. These are mostly assumptions that remain to be elucidated in future studies.

Contrary to the expected outcome that provinces with higher health expenditures would exhibit a reduced risk of temperature-related dengue risk due to the potential focus of these resources on overall health enhancement, our findings indicate a higher temperature-related dengue risk in areas with greater health spending as opposed to those with lower health expenditures The increased dengue risk amidst high health spending may seem counterintuitive, however, health spending does not necessarily equate to improved health. Dieleman et. al. [66] conducted a study analyzing factors associated with increases in US health care spending from 1996 to 2013. The study found that despite the substantial increase in health care spending during this period, there was no significant improvement in health outcomes, such as mortality rates or disease burden. The linkage between health care spending and improvement of health outcome is not direct and may be mediated by several factors. In Sub-Saharan Africa (SSA), Makuta and O'Hare [67] examined the relationship between public spending on health, quality of governance, and health outcomes. Their findings indicated that health spending has a statistically significant impact on improving health outcomes, such as under-five mortality and life expectancy. However, the impact of health spending on health outcomes is mediated by the quality of governance. Grigoli and Kapsoli [68] also emphasized the need to consider the efficiency of health spending. Simply increasing public expenditure in the health sector may not significantly affect health outcomes if the efficiency of spending is low. Therefore, it is crucial to ensure that health spending is allocated and utilized efficiently to achieve desired improvements in health outcomes. Similarly, the increased in temperature-related dengue risk may also be due to the higher health-seeking behavior of the population, especially in resource-rich locations. Consistent with a previous literature [69], hydrometeorological changes were found to substantially affect the risk of dengue in regions with mid-to-low latitudes in China. Patz et. al. [8] assert that increased incidence of dengue may first occur in regions bordering endemic zones in latitude or altitude. In summary, from a programmatic, local risk standpoint, our study shows that the association can be modified by area specific factors, thus merits the need for a tailor-fit dengue-control response. The Philippines may benefit from devolved decentralized and localized dengue control and early warning systems in order to account for these variabilities.

Limitations

The current study has several limitations. Firstly, there is limited availability, accessibility, and data quality of surveillance data for dengue cases and meteorological variables for all provinces. The available dengue case data set is constrained by varied case definitions and underreported dengue cases, as only clinically apparent cases were recorded. Locations with higher poverty incidence may be less likely to seek healthcare assistance due to less knowledge, accessibility, or financial capacity, resulting in lower reported dengue cases, and thus, inaccurately minimizing the modifying effect of poverty on the temperature-dengue relationship. These data sets were also subjected to varied implementation measures per province, making it difficult to determine if increased cases were related to improved surveillance or increases in

regional or global temperatures. Secondly, the absence of a historical time series of socio-economic variables has led us to treat data from 2015 as a representation of the data from 2010–2019. There is also a lack of complete, provincial data for other possibly modifying variables, such as forest cover and the presence of vegetation, open water containers, and degree of intra-household crowding. Future studies may benefit and have an improved understanding of dengue modeling if these additional spatial and socio-economic factors are included. Relationships between components of the model and the chosen socio-economic variables may change over time in ways that are difficult to predict, highlighting the need for more granular, long-term surveillance data.

Conclusions

This observational study found that temperature was associated with dengue incidence, and that this association is more apparent in locations with high population density, less people in urban areas with larger household size, high poverty incidence, higher health spending per capita, and in lower latitudes. Differences with socio-economic conditions is linked with dengue risk. This highlights the need to develop interventions tailor-fit to local conditions.

Consent for publication

Not applicable.

Supporting information

S1 Table. Sources of data meta-regressors.
(DOCX)

S2 Table. Description of socio-economic factors used for the study. The socio-economic factors used were province specific. All were originally yearly data, except for latitude. The corresponding median values of the socio-economic factors were used, depending on the distribution of each dataset.
(DOCX)

S3 Table. Main effects of the effect modifiers.
(DOCX)

S1 Fig. Correlation plot between the temperature data from ERA5-land and existing background monitoring stations in cities of Manila and Cebu.
(DOCX)

S2 Fig. Maximum lag sensitivity analyses.
(DOCX)

S3 Fig. Sensitivity analysis used for df selection.
(DOCX)

S4 Fig. Variation in the annual distribution of cases per year.
(DOCX)

S5 Fig. Sensitivity analyses adjusting for precipitation derived from ERA5-land.
(DOCX)

S6 Fig. Temperature-specific associations in different temperature percentiles.
(DOCX)

S7 Fig. Interaction analyses per effect modifier.
(DOCX)

Acknowledgments

We acknowledge the input from the Philippine Atmospheric, Geophysical, and Astronomical Services Administration (PAGASA) and data from the Philippine Statistics Authority (PSA).

Author Contributions

Conceptualization: Xerxes Seposo, Sary Valenzuela, Geminn Louis Apostol.

Data curation: Xerxes Seposo, Sary Valenzuela, Geminn Louis Apostol.

Formal analysis: Xerxes Seposo.

Funding acquisition: Xerxes Seposo, Sary Valenzuela, Geminn Louis Apostol.

Investigation: Xerxes Seposo.

Methodology: Xerxes Seposo.

Project administration: Xerxes Seposo, Sary Valenzuela, Geminn Louis Apostol.

Resources: Sary Valenzuela, Geminn Louis Apostol.

Software: Xerxes Seposo.

Visualization: Xerxes Seposo.

Writing – original draft: Xerxes Seposo, Sary Valenzuela, Geminn Louis Apostol.

Writing – review & editing: Xerxes Seposo, Sary Valenzuela, Geminn Louis Apostol.

References

1. Naish S, Dale P, Mackenzie JS, McBride J, Mengersen K, Tong S. Climate change and dengue: a critical and systematic review of quantitative modelling approaches. BMC Infect Dis. 2014; 14:167. https://doi.org/10.1186/1471-2334-14-167 PMID: 24669859
2. WHO. Dengue and severe dengue. World Health Organization; 2023.
3. WHO-WPRO. Dengue situation updates 2019. Manila, Philippines: WHO-WPRO; 2020.
4. Morin CW, Comrie AC, Ernst K. Climate and dengue transmission: evidence and implications. Environ Health Perspect. 2013; 121(11–12):1264–72. https://doi.org/10.1289/ehp.1306556 PMID: 24058050
5. Li Y, Dou Q, Lu Y, Xiang H, Yu X, Liu S. Effects of ambient temperature and precipitation on the risk of dengue fever: A systematic review and updated meta-analysis. Environ Res. 2020; 191:110043. https://doi.org/10.1016/j.envres.2020.110043 PMID: 32810500
6. Fan J, Wei W, Bai Z, Fan C, Li S, Liu Q, et al. A systematic review and meta-analysis of dengue risk with temperature change. Int J Environ Res Public Health. 2014; 12(1):1–15. https://doi.org/10.3390/ijerph120100001 PMID: 25546270
7. Jetten TH, Focks DA. Potential changes in the distribution of dengue transmission under climate warming. Am J Trop Med Hyg. 1997; 57(3):285–97. https://doi.org/10.4269/ajtmh.1997.57.285 PMID: 9311638
8. Patz JA, Martens WJ, Focks DA, Jetten TH. Dengue fever epidemic potential as projected by general circulation models of global climate change. Environmental Health Perspectives. 1998; 106(3):147–53. https://doi.org/10.1289/ehp.98106147 PMID: 9452414
9. Servadio JL, Rosenthal SR, Carlson L, Bauer C. Climate patterns and mosquito-borne disease outbreaks in South and Southeast Asia. J Infect Public Health. 2018; 11(4):566–71. https://doi.org/10.1016/j.jiph.2017.12.006 PMID: 29274851
10. Watts DM, Burke DS, Harrison BA, Whitmire RE, Nisalak A. Effect of temperature on the vector efficiency of Aedes aegypti for dengue 2 virus. Am J Trop Med Hyg. 1987; 36(1):143–52. https://doi.org/10.4269/ajtmh.1987.36.143 PMID: 3812879

11. Arcari P, Tapper N, Pfueller S. Regional variability in relationships between climate and dengue/DHF in Indonesia. Singapore Journal of Tropical Geography. 2007; 28(3):251–72.
12. Barbazan P, Guiserix M, Boonyuan W, Tuntaprasart W, Pontier D, Gonzalez JP. Modelling the effect of temperature on transmission of dengue. Med Vet Entomol. 2010; 24(1):66–73. https://doi.org/10.1111/j.1365-2915.2009.00848.x PMID: 20377733
13. Hii YL, Rocklov J, Ng N, Tang CS, Pang FY, Sauerborn R. Climate variability and increase in intensity and magnitude of dengue incidence in Singapore. Glob Health Action. 2009;2. https://doi.org/10.3402/gha.v2i0.2036 PMID: 20052380
14. Wu P-C, Guo H-R, Lung S-C, Lin C-Y, Su H-J. Weather as an effective predictor for occurrence of dengue fever in Taiwan. Acta Tropica. 2007; 103(1):50–7. https://doi.org/10.1016/j.actatropica.2007.05.014 PMID: 17612499
15. Edillo F, Ymbong RR, Bolneo AA, Hernandez RJ, Fuentes BL, Cortes G, et al. Temperature, season, and latitude influence development-related phenotypes of Philippine Aedes aegypti (Linnaeus): Implications for dengue control amidst global warming. Parasit Vectors. 2022; 15(1):74. https://doi.org/10.1186/s13071-022-05186-x PMID: 35248140
16. Teurlai M, Menkes CE, Cavarero V, Degallier N, Descloux E, Grangeon JP, et al. Socio-economic and Climate Factors Associated with Dengue Fever Spatial Heterogeneity: A Worked Example in New Caledonia. PLoS Negl Trop Dis. 2015; 9(12):e0004211. https://doi.org/10.1371/journal.pntd.0004211 PMID: 26624008
17. Vincenti-Gonzalez MF, Grillet ME, Velasco-Salas ZI, Lizarazo EF, Amarista MA, Sierra GM, et al. Spatial Analysis of Dengue Seroprevalence and Modeling of Transmission Risk Factors in a Dengue Hyperendemic City of Venezuela. PLoS Negl Trop Dis. 2017; 11(1):e0005317. https://doi.org/10.1371/journal.pntd.0005317 PMID: 28114342
18. Watts MJ, Kotsila P, Mortyn PG, Sarto IMV, Urzi Brancati C. Influence of socio-economic, demographic and climate factors on the regional distribution of dengue in the United States and Mexico. Int J Health Geogr. 2020; 19(1):44. https://doi.org/10.1186/s12942-020-00241-1 PMID: 33138827
19. Zheng L, Ren HY, Shi RH, Lu L. Spatiotemporal characteristics and primary influencing factors of typical dengue fever epidemics in China. Infect Dis Poverty. 2019; 8(1):24. https://doi.org/10.1186/s40249-019-0533-9 PMID: 30922405
20. da Conceição Araújo D, dos Santos AD, Lima SVMA, Vaez AC, Cunha JO, Conceição Gomes Machado de Araújo K. Determining the association between dengue and social inequality factors in north-eastern Brazil: A spatial modelling. Geospatial Health. 2020; 15(1).
21. Farinelli EC, Baquero OS, Stephan C, Chiaravalloti-Neto F. Low socioeconomic condition and the risk of dengue fever: A direct relationship. Acta Trop. 2018; 180:47–57. https://doi.org/10.1016/j.actatropica.2018.01.005 PMID: 29352990
22. Haakenstad A, Coates M, Bukhman G, McConnell M, Verguet S. Comparative health systems analysis of differences in the catastrophic health expenditure associated with non-communicable vs communicable diseases among adults in six countries. Health Policy and Planning. 2022; 37(9):1107–15. https://doi.org/10.1093/heapol/czac053 PMID: 35819006
23. Muñoz-Sabater J, Dutra E, Agustí-Panareda A, Albergel C, Arduini G, Balsamo G, et al. ERA5-Land: a state-of-the-art global reanalysis dataset for land applications. Earth Syst Sci Data. 2021; 13(9):4349–83.
24. Yilmaz M. Accuracy assessment of temperature trends from ERA5 and ERA5-Land. Sci Total Environ. 2023; 856(Pt 2):159182. https://doi.org/10.1016/j.scitotenv.2022.159182 PMID: 36195144
25. Zou J, Lu N, Jiang H, Qin J, Yao L, Xin Y, et al. Performance of air temperature from ERA5-Land reanalysis in coastal urban agglomeration of Southeast China. Sci Total Environ. 2022; 828:154459. https://doi.org/10.1016/j.scitotenv.2022.154459 PMID: 35278562
26. He C, Kim H, Hashizume M, Lee W, Honda Y, Kim SE, et al. The effects of night-time warming on mortality burden under future climate change scenarios: a modelling study. The Lancet Planetary Health. 2022; 6(8):e648–e57. https://doi.org/10.1016/S2542-5196(22)00139-5 PMID: 35932785
27. Mistry MN, Schneider R, Masselot P, Royé D, Armstrong B, Kyselý J, et al. Comparison of weather station and climate reanalysis data for modelling temperature-related mortality. Scientific Reports. 2022; 12(1):5178. https://doi.org/10.1038/s41598-022-09049-4 PMID: 35338191
28. Urban A, Di Napoli C, Cloke HL, Kyselý J, Pappenberger F, Sera F, et al. Evaluation of the ERA5 reanalysis-based Universal Thermal Climate Index on mortality data in Europe. Environmental Research. 2021; 198:111227. https://doi.org/10.1016/j.envres.2021.111227 PMID: 33974842
29. Bavia L, Melanda FN, de Arruda TB, Mosimann ALP, Silveira GF, Aoki MN, et al. Epidemiological study on dengue in southern Brazil under the perspective of climate and poverty. Sci Rep. 2020; 10(1):2127. https://doi.org/10.1038/s41598-020-58542-1 PMID: 32034173

30. Castro M, Mattos E, Patriota F. The effects of health spending on the propagation of infectious diseases. Health Econ. 2021; 30(10):2323–44. https://doi.org/10.1002/hec.4388 PMID: 34247434

31. do Carmo RF, Silva Júnior JVJ, Pastor AF, de Souza CDF. Spatiotemporal dynamics, risk areas and social determinants of dengue in Northeastern Brazil, 2014–2017: an ecological study. Infectious Diseases of Poverty. 2020; 9(1):153. https://doi.org/10.1186/s40249-020-00772-6 PMID: 33143752

32. Mutucumarana CP, Bodinayake CK, Nagahawatte A, Devasiri V, Kurukulasooriya R, Anuradha T, et al. Geospatial analysis of dengue emergence in rural areas in the Southern Province of Sri Lanka. Trans R Soc Trop Med Hyg. 2020; 114(6):408–14. https://doi.org/10.1093/trstmh/trz123 PMID: 31885050

33. Romeo-Aznar V, Picinini Freitas L, Goncalves Cruz O, King AA, Pascual M. Author Correction: Fine-scale heterogeneity in population density predicts wave dynamics in dengue epidemics. Nat Commun. 2022; 13(1):1404. https://doi.org/10.1038/s41467-022-29163-1 PMID: 35277517

34. Schmidt WP, Suzuki M, Thiem VD, White RG, Tsuzuki A, Yoshida LM, et al. Population density, water supply, and the risk of dengue fever in Vietnam: cohort study and spatial analysis. PLoS Med. 2011; 8 (8):e1001082. https://doi.org/10.1371/journal.pmed.1001082 PMID: 21918642

35. Colon-Gonzalez FJ, Sewe MO, Tompkins AM, Sjodin H, Casallas A, Rocklov J, et al. Projecting the risk of mosquito-borne diseases in a warmer and more populated world: a multi-model, multi-scenario intercomparison modelling study. Lancet Planet Health. 2021; 5(7):e404–e14. https://doi.org/10.1016/S2542-5196(21)00132-7 PMID: 34245711

36. Hii YL, Rocklov J, Wall S, Ng LC, Tang CS, Ng N. Optimal lead time for dengue forecast. PLoS Negl Trop Dis. 2012; 6(10):e1848. https://doi.org/10.1371/journal.pntd.0001848 PMID: 23110242

37. Imai C, Armstrong B, Chalabi Z, Mangtani P, Hashizume M. Time series regression model for infectious disease and weather. Environ Res. 2015; 142:319–27. https://doi.org/10.1016/j.envres.2015.06.040 PMID: 26188633

38. Gasparrini A, Guo Y, Hashizume M, Kinney PL, Petkova EP, Lavigne E, et al. Temporal Variation in Heat–Mortality Associations: A Multicountry Study. Environmental Health Perspectives. 2015; 123 (11):1200–7. https://doi.org/10.1289/ehp.1409070 PMID: 25933359

39. Gasparrini A. Distributed Lag Linear and Non-Linear Models in R: The Package dlnm. J Stat Softw. 2011; 43(8):1–20. PMID: 22003319

40. Sera F, Armstrong B, Blangiardo M, Gasparrini A. An extended mixed-effects framework for meta-analysis. Stat Med. 2019; 38(29):5429–44. https://doi.org/10.1002/sim.8362 PMID: 31647135

41. Wood SN. Fast stable restricted maximum likelihood and marginal likelihood estimation of semiparametric generalized linear models. Journal of the Royal Statistical Society: Series B (Statistical Methodology). 2011; 73(1):3–36.

42. Grolemund G, Wickham H. Dates and Times Made Easy with lubridate. Journal of Statistical Software. 2011; 40(3):1–25.

43. Wickham H, Francois R, Henry L, K. M. dplyr: A Grammar of Data Manipulation 2018 [Available from: https://CRAN. R-project. org/package = dplyr.

44. Strachan AL. Conflict Analysis of Muslim Mindanao. 2015.

45. PSA. Highlights of the Philippine Population 2020 Census of Population and Housing (2020 CPH). Manila, Philippines: Philippine Statistics Agency; 2021.

46. Xiang J, Hansen A, Liu Q, Liu X, Tong MX, Sun Y, et al. Association between dengue fever incidence and meteorological factors in Guangzhou, China, 2005–2014. Environmental Research. 2017; 153:17–26. https://doi.org/10.1016/j.envres.2016.11.009 PMID: 27883970

47. Lambrechts L, Paaijmans KP, Fansiri T, Carrington LB, Kramer LD, Thomas MB, et al. Impact of daily temperature fluctuations on dengue virus transmission by Aedes aegypti. Proc Natl Acad Sci U S A. 2011; 108(18):7460–5. https://doi.org/10.1073/pnas.1101377108 PMID: 21502510

48. Wang Y, Wei Y, Li K, Jiang X, Li C, Yue Q, et al. Impact of extreme weather on dengue fever infection in four Asian countries: A modelling analysis. Environ Int. 2022; 169:107518. https://doi.org/10.1016/j.envint.2022.107518 PMID: 36155913

49. Hawley WA. The biology of Aedes albopictus. Journal of the American Mosquito Control Association Supplement. 1988; 1:1–39. PMID: 3068349

50. Delatte H, Gimonneau G, Triboire A, Fontenille D. Influence of temperature on immature development, survival, longevity, fecundity, and gonotrophic cycles of Aedes albopictus, vector of chikungunya and dengue in the Indian Ocean. J Med Entomol. 2009; 46(1):33–41. https://doi.org/10.1603/033.046.0105 PMID: 19198515

51. Gui H, Gwee S, Koh J, Pang J. Weather Factors Associated with Reduced Risk of Dengue Transmission in an Urbanized Tropical City. Int J Environ Res Public Health. 2021; 19(1).

52. Koh BK, Ng LC, Kita Y, Tang CS, Ang LW, Wong KY, et al. The 2005 dengue epidemic in Singapore: epidemiology, prevention and control. Ann Acad Med Singap. 2008; 37(7):538–45. PMID: 18695764

53. Heng BH, Goh KT, Neo KS, editors. Environmental temperature, Aedes aegypti house index and rainfall as predictors of annual epidemics of dengue fever and dengue haemorrhagic fever in Singapore1998.

54. Polwiang S. The time series seasonal patterns of dengue fever and associated weather variables in Bangkok (2003–2017). BMC Infect Dis. 2020; 20(1):208. https://doi.org/10.1186/s12879-020-4902-6 PMID: 32164548

55. Sharmin S, Glass K, Viennet E, Harley D. Interaction of Mean Temperature and Daily Fluctuation Influences Dengue Incidence in Dhaka, Bangladesh. PLOS Neglected Tropical Diseases. 2015; 9(7): e0003901. https://doi.org/10.1371/journal.pntd.0003901 PMID: 26161895

56. Nuranisa R, Budi Maryanto Y, Atoillah Isfandiari M. CORRELATION OF FREE LARVAE INDEX AND POPULATION DENSITY WITH DENGUE FEVER INCIDENCE RATE. The Indonesian Journal Of Public Health 2022; 17(3).

57. Bhatt S, Gething PW, Brady OJ, Messina JP, Farlow AW, Moyes CL, et al. The global distribution and burden of dengue. Nature. 2013; 496(7446):504–7. https://doi.org/10.1038/nature12060 PMID: 23563266

58. Braga C, Luna CF, Martelli CM, de Souza WV, Cordeiro MT, Alexander N, et al. Seroprevalence and risk factors for dengue infection in socio-economically distinct areas of Recife, Brazil. Acta Trop. 2010; 113(3):234–40. https://doi.org/10.1016/j.actatropica.2009.10.021 PMID: 19896921

59. Velasco-Salas ZI, Sierra GM, Guzman DM, Zambrano J, Vivas D, Comach G, et al. Dengue seroprevalence and risk factors for past and recent viral transmission in Venezuela: a comprehensive community-based study. Am J Trop Med Hyg. 2014; 91(5):1039–48. https://doi.org/10.4269/ajtmh.14-0127 PMID: 25223944

60. Chang AY, Fuller DO, Carrasquillo O, Beier JC. Social justice, climate change, and dengue. Health Hum Rights. 2014; 16(1):93–104. PMID: 25474614

61. Mulligan K, Dixon J, Joanna Sinn C-L, Elliott SJ. Is dengue a disease of poverty? A systematic review. Pathogens and Global Health. 2015; 109(1):10–8. https://doi.org/10.1179/2047773214Y.0000000168 PMID: 25546339

62. Freitas MGR, Tsouris P, Reis ICd, Magalhães MdAFM, Nascimento TFSd, Rocha NH. DENGUE AND LAND COVER HETEROGENEITY IN RIO DE JANEIRO. Oecologia Australis. 2010; 14(3):641–67.

63. Pangilinan MAP, Gonzales DPG, Leong RNF, Co FF. Spatial Analysis of the Distribution of Reported Dengue Incidence in the National Capital Region, Philippines. Acta Medica Philippina. 2017; 51(2).

64. Marigmen JLDC, Balino LVA, Addawe RC. Analysis of dengue incidence in Baguio, Philippines. AIP Conference Proceedings. 2021; 2423(1).

65. Waterman SH, Novak RJ, Sather GE, Bailey RE, Rios I, Gubler DJ. Dengue transmission in two Puerto Rican communities in 1982. Am J Trop Med Hyg. 1985; 34(3):625–32. https://doi.org/10.4269/ajtmh.1985.34.625 PMID: 4003671

66. Dieleman JL, Squires E, Bui AL, Campbell M, Chapin A, Hamavid H, et al. Factors Associated With Increases in US Health Care Spending, 1996–2013. JAMA. 2017; 318(17):1668–78. https://doi.org/10.1001/jama.2017.15927 PMID: 29114831

67. Makuta I, O'Hare B. Quality of governance, public spending on health and health status in Sub Saharan Africa: a panel data regression analysis. BMC Public Health. 2015; 15(1):932. https://doi.org/10.1186/s12889-015-2287-z PMID: 26390867

68. Grigoli F, Kapsoli J. Waste not, want not: The efficiency of health expenditure in emerging and developing economies. Review of Development Economics. 2018; 22(1):384–403.

69. Li C, Liu Z, Li W, Lin Y, Hou L, Niu S, et al. Projecting future risk of dengue related to hydrometeorological conditions in mainland China under climate change scenarios: a modelling study. The Lancet Planetary Health. 2023; 7(5):e397–e406. https://doi.org/10.1016/S2542-5196(23)00051-7 PMID: 37164516